



Semester I Examinations 2016/17

Exam Code(s)	1SD1, 1SD3, 1MF1
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Liam Maguire
Internal Examiner(s)	Dr. Jim Duggan *Dr. Michael Schukat

Instructions: Answer two questions from Section A and one question from Section B.

Duration	2 hrs
No. of Pages	4
Discipline(s)	Information Technology

Requirements	None
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Release to Library: Yes ☒ No ☐

Section A

Question 1

- a) Using examples distinguish between *RISC instructions* and *CISC instructions*. Briefly discuss what type of CPU (e.g. RISC or CISC) can be typically found in small battery operated computing devices, e.g. smartphones. Explain your answer.
8 MARKS
- b) Explain what is meant by DMA (Direct Memory Access) and how it is used in a computer?
6 MARKS
- c) Distinguish between the two machine instruction types *conditional jump* and *unconditional jump* (i.e. mnemonics JMP and JNZ as discussed in lectures). Show how both instruction types can be used to implement an “if-then-else” block.
8 MARKS
- a) Explain the inner workings of a hard disk with multiple platters. In your answer refer to the terms *sector*, *track* and *cylinder*.
8 MARKS

Question 2

- a) Using *two-complement binary representation* evaluate the following mathematical expressions:
3 + 4
(-5) – 3
6 + (-1)
All numbers should be represented as bytes.
9 MARKS
- b) Explain *hexadecimal representation* and outline its benefits.
5 MARKS
- c) What is meant by a *numeric overflow* in binary addition / subtraction, and how can it be detected?
6 MARKS
- d) Using examples distinguish between the following process scheduling algorithms
- shortest process next
- shortest remaining time
- round robin.
10 MARKS

PTO

Question 3

- a) Using examples distinguish between the following secondary storage management techniques
- contiguous allocation
 - chained allocation.

6 MARKS

- b) Using an example explain the purpose of the *address lines*, *data lines* and *control lines* of a bus.

8 MARKS

- c) What is an *interrupt* and what is meant by the term *interrupt processing*? In your answer refer to the terms *interrupt priority*, *interrupt service routine* and *nested interrupts*.

8 MARKS

- d) Summarise the 7-state process model and, using an example, show how a process moves through all different states.

8 MARKS

PTO

Section B

Question 4

- a) Distinguish between the protocols TCP and IP. In your answer discuss content and structure of the headers of both protocols.

10 MARKS

- b) Discuss similarities and differences between the multiple access protocols *pure ALOHA* and *Carrier Sense Multiple Access (CSMA)*. Explain which of the 2 protocols generally performs better.

8 MARKS

- c) Briefly summarise similarities and differences between a *repeater*, a *hub* and a *switch*.

6 MARKS

- a) *Network address translation* (NAT) is a common feature of modern network routers. Explain how NAT works and how it can protect a network from unsolicited external access.

6 MARKS

Question 5

- a) Using examples show how TCP uses a 3-way handshake to establish and to release a connection.

9 MARKS

- b) *Network address translation* (NAT) is a common feature of modern network routers. Explain how NAT works and how it can protect a network from unsolicited external access.

6 MARKS

- c) Using an example show how the *shortest path routing algorithm* works.

10 MARKS

- d) Briefly summarise *Nagle's Algorithm* and discuss its advantages and disadvantages.

5 MARKS